

Introduction to Abstract Algebra III: Homework 2

Due on April 15, 2026 at 23:59

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Exercise 40.6. Find the degree and a basis for the field extension $\mathbb{Q}(\sqrt{2} + \sqrt{3})$ over $\mathbb{Q}$.

Solution. The basis for the field extension $\mathbb{Q}(\sqrt{2} + \sqrt{3})$ over $\mathbb{Q}$ is $\{1, \sqrt{2} + \sqrt{3}, (\sqrt{2} + \sqrt{3})^2, (\sqrt{2} + \sqrt{3})^3\}$. The degree of the field extension is 4, since the minimal polynomial of $\sqrt{2} + \sqrt{3}$ over $\mathbb{Q}$ is given by $f(x) = x^4 - 10x + 1$. This polynomial is irreducible over $\mathbb{Q}$ by Eisenstein's criterion with $p = 5$, and thus the degree of the extension is equal to the degree of the minimal polynomial, which is 4. $\square$

Exercise 40.8. Find the degree and a basis for the field extension $\mathbb{Q}(\sqrt{2}, \sqrt[3]{5})$ over $\mathbb{Q}$.

Solution. The basis for the field extension $\mathbb{Q}(\sqrt{2}, \sqrt[3]{5})$ over $\mathbb{Q}$ is $\{1, \sqrt{2}, \sqrt[3]{5}, \sqrt{2}\sqrt[3]{5}, (\sqrt[3]{5})^2, \sqrt{2}(\sqrt[3]{5})^2\}$. The degree of the field extension is 6, since the minimal polynomial of $\sqrt{2}$ over $\mathbb{Q}$ is $x^2 - 2$, which has degree 2, and the minimal polynomial of $\sqrt[3]{5}$ over $\mathbb{Q}$ is $x^3 - 5$, which has degree 3. Since these two extensions are linearly disjoint, the degree of the combined extension is the product of the degrees, which is $2 \cdot 3 = 6$. $\square$

Exercise 40.10. Find the degree and a basis for the field extension $\mathbb{Q}(\sqrt{2}, \sqrt{6})$ over $\mathbb{Q}(\sqrt{3})$.

Solution. Note that $\sqrt{6}/\sqrt{2} = \sqrt{3}$. Thus, $\mathbb{Q}(\sqrt{2}, \sqrt{6}) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$. This problem is now reduced to finding the basis and degree of the field extension $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ over $\mathbb{Q}(\sqrt{3})$. The basis for this extension is $\{1, \sqrt{2}\}$, since $\sqrt{2}$ is not in $\mathbb{Q}(\sqrt{3})$. The degree of the field extension is 2, since the minimal polynomial of $\sqrt{2}$ over $\mathbb{Q}(\sqrt{3})$ is $x^2 - 2$, which has degree 2. Thus, the degree of the field extension $\mathbb{Q}(\sqrt{2}, \sqrt{6})$ over $\mathbb{Q}(\sqrt{3})$ is 2, and a basis for this extension is $\{1, \sqrt{2}\}$. $\square$

Exercise 40.18. Show by an example that for a proper extension field E of a field F , the algebraic closure of F in E need not be algebraically closed.

Solution. Consider the field extension $\mathbb{Q}(\sqrt{2})$ over $\mathbb{Q}$. The algebraic closure of $\mathbb{Q}$ in $\mathbb{Q}(\sqrt{2})$ is $\mathbb{Q}$, which is not algebraically closed since it does not contain all the roots of polynomials with coefficients in $\mathbb{Q}$. For example, the polynomial $x^2 - 2$ has a root $\sqrt{2}$ in $\mathbb{Q}(\sqrt{2})$, but it does not have any roots in $\mathbb{Q}$. Thus, the algebraic closure of $\mathbb{Q}$ in $\mathbb{Q}(\sqrt{2})$ is not algebraically closed. $\square$

Exercise 40.23. Show that if E is a finite extension of a field F and $[E : F]$ is a prime number, then E is a simple extension of F and, indeed, $E = F(\alpha)$ for every $\alpha \in E$ not in F .

Solution. Since α is algebraic over F , there exists a minimal polynomial $f(x) \in F[x]$ such that $f(\alpha) = 0$. The degree of this minimal polynomial divides the degree of the extension, which is p . Since p is prime, the degree of the minimal polynomial must be either 1 or p . If the degree is 1, then $\alpha \in F$, which contradicts our assumption. Therefore, the degree of the minimal polynomial must be p , and thus $[F(\alpha) : F] = p$.

Since $[E : F] = p$ and $[F(\alpha) : F] = p$, we have that $E = F(\alpha)$, as there can be no intermediate fields between F and E . Therefore, every element $\alpha \in E$ that is not in F generates the entire extension, and we conclude that E is a simple extension of F . $\square$

Exercise 40.24. Prove that $x^2 - 3$ is irreducible over $\mathbb{Q}(\sqrt[3]{2})$.

Solution. We know that $[\mathbb{Q}(\sqrt{3}) : \mathbb{Q}] = 2$ and $[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3$. Now, suppose for contradiction, that $\sqrt{3} \in \mathbb{Q}(\sqrt[3]{2})$. Then, $\mathbb{Q}(\sqrt{3}) \subseteq \mathbb{Q}(\sqrt[3]{2})$. Then, we have

$$[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = [\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}(\sqrt{3})][\mathbb{Q}(\sqrt{3}) : \mathbb{Q}].$$

So, we have

$$3 = [\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}(\sqrt{3})] \cdot 2,$$

which is a contradiction since the left-hand side is odd and the right-hand side is even. Thus, $\sqrt{3} \notin \mathbb{Q}(\sqrt[3]{2})$, and therefore $x^2 - 3$ is irreducible over $\mathbb{Q}(\sqrt[3]{2})$. $\square$

Exercise 40.25. What degree field extensions can we obtain by successively adjoining to a field F a square root of an element of F not a square in F , then a square root of some nonsquare in this new field, and so on? Argue from this that a zero of $x^{14} - 3x^2 + 12$ over $\mathbb{Q}$ can never be expressed as a rational function of square roots of rational functions of square roots, and so on, of elements of $\mathbb{Q}$.

Solution. By successively adjoining square roots of nonsquares, we can only obtain field extensions of degree 2^n for some nonnegative integer n . This is because each time we adjoin a square root, we either get a degree 1 extension (if the element is already a square) or a degree 2 extension (if the element is not a square). Therefore, the degree of the resulting field extension after n steps will be at most 2^n .

Now, consider the polynomial $x^{14} - 3x^2 + 12$. The degree of this polynomial is 14, which is not a power of 2. Therefore, any field extension that contains a zero of this polynomial must have degree that divides 14. Since the only powers of 2 that divide 14 are 1 and 2, it follows that any field extension containing a zero of this polynomial cannot be obtained by successively adjoining square roots as described. Thus, a zero of $x^{14} - 3x^2 + 12$ over $\mathbb{Q}$ can never be expressed as a rational function of square roots of rational functions of square roots, and so on, of elements of $\mathbb{Q}$. $\square$

Exercise 40.28. Generalizing Exercise 27, show that if $\sqrt{a} + \sqrt{b} \neq 0$, then $\mathbb{Q}(\sqrt{a} + \sqrt{b}) = \mathbb{Q}(\sqrt{a}, \sqrt{b})$ for all a and b in $\mathbb{Q}$. [Hint: Compute $(a - b)/(\sqrt{a} + \sqrt{b})$.]

Solution. Computing $(a - b)/(\sqrt{a} + \sqrt{b})$, we have

$$\frac{a - b}{\sqrt{a} + \sqrt{b}} = \frac{(a - b)(\sqrt{a} - \sqrt{b})}{(\sqrt{a} + \sqrt{b})(\sqrt{a} - \sqrt{b})} = \frac{(a - b)(\sqrt{a} - \sqrt{b})}{a - b} = \sqrt{a} - \sqrt{b}.$$

Thus, we have $\sqrt{a} - \sqrt{b} \in \mathbb{Q}(\sqrt{a} + \sqrt{b})$. Since $\sqrt{a} + \sqrt{b} \in \mathbb{Q}(\sqrt{a} + \sqrt{b})$, we have $\sqrt{a} = (\sqrt{a} + \sqrt{b} + \sqrt{a} - \sqrt{b})/2$ and $\sqrt{b} = (\sqrt{a} + \sqrt{b} - (\sqrt{a} - \sqrt{b}))/2$ are also in $\mathbb{Q}(\sqrt{a} + \sqrt{b})$. Thus, we have $\mathbb{Q}(\sqrt{a}, \sqrt{b}) \subseteq \mathbb{Q}(\sqrt{a} + \sqrt{b})$. The other inclusion is trivial since $\mathbb{Q}(\sqrt{a} + \sqrt{b})$ is generated by $\sqrt{a}$ and $\sqrt{b}$, which are in $\mathbb{Q}(\sqrt{a}, \sqrt{b})$. Therefore, we conclude that $\mathbb{Q}(\sqrt{a} + \sqrt{b}) = \mathbb{Q}(\sqrt{a}, \sqrt{b})$. $\square$

Exercise 40.29. Let E be a finite extension of a field F , and let $p(x) \in F[x]$ be irreducible over F and have degree that is not a divisor of $[E : F]$. Show that $p(x)$ has no zeros in E .

Solution. Suppose for contradiction that $p(x)$ has a zero α in E . Then, we have the field extension $F \subseteq F(\alpha) \subseteq E$. Since $p(x)$ is irreducible over F , the degree of the extension $[F(\alpha) : F]$ is equal to the degree of $p(x)$, which we denote as n . Thus, we have

$$[E : F] = [E : F(\alpha)][F(\alpha) : F] = [E : F(\alpha)]n.$$

This implies that n divides $[E : F]$, which contradicts our assumption. Therefore, we conclude that $p(x)$ has no zeros in E . $\square$

Exercise 40.30. Let E be an extension field of F . Let $\alpha \in E$ be algebraic of odd degree over F . Show that α^2 is algebraic of odd degree over F , and $F(\alpha) = F(\alpha^2)$.

Solution. We first show that if α is algebraic of odd degree over F , then α^2 is also algebraic of odd degree over F . Since α^2 is a power of α , we have the inclusion property $F \subseteq F(\alpha^2) \subseteq F(\alpha)$. Thus, we have

$$[F(\alpha) : F] = [F(\alpha) : F(\alpha^2)][F(\alpha^2) : F].$$

We are given that $[F(\alpha) : F] = n$, for some odd integer n . Consider the polynomial $f(x) = x^2 - \alpha^2 \in F(\alpha^2)[x]$. Since α is a root of $f(x)$, the minimal polynomial of α must divide $x^2 - \alpha^2$. Thus, we have $[F(\alpha) : F(\alpha^2)]$ can only be 1 or 2. If it is 1, then $F(\alpha) = F(\alpha^2)$, and we are done. If it is 2, then we have

$$n = [F(\alpha) : F] = [F(\alpha) : F(\alpha^2)][F(\alpha^2) : F] = 2[F(\alpha^2) : F].$$

This is, however, a contradiction since n is odd. Thus, we must have $[F(\alpha) : F(\alpha^2)] = 1$, and therefore $F(\alpha) = F(\alpha^2)$. $\square$